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AI as a Behavioral Signal: How LLM Reveals High-Value Finance User Applications

Thomas Kim, Cheeseng Her, Siyu Liang, and Shreya Muthukumar

User Behavior Shifts in the AI-Driven Mobile App Ecosystem

Understanding these shifts is critical for businesses to adapt to changing user behavior, remain competitive, and identify high-value user segments in the evolving mobile app ecosystem.

1

Rise of Generative AI

Generative AI tools like ChatGPT, Gemini, and Perplexity are rapidly growing and becoming part of daily digital behavior, especially among students and digital-native users.

2

Co-Usage with LLM Apps

Our project examines which non-LLM apps are used alongside LLM tools and identifies the strongest co-usage patterns that characterize AI adoption behavior.

3

Business Insight

Understanding these patterns helps businesses identify user segments, uncover behavior trends, and develop strategies to better engage high-value users in a changing app ecosystem. In particular, we focus on whether AI adoption signals high-intent behaviors such as financial app usage.

SITUATION

Generative AI tools such as ChatGPT, Gemini, and Perplexity are rapidly growing and becoming integrated into users' daily digital behavior.

COMPLICATION

It is unclear how AI adoption affects usage of other apps, creating uncertainty around whether LLM tools replace, reduce, or complement existing app engagement.

KEY QUESTION

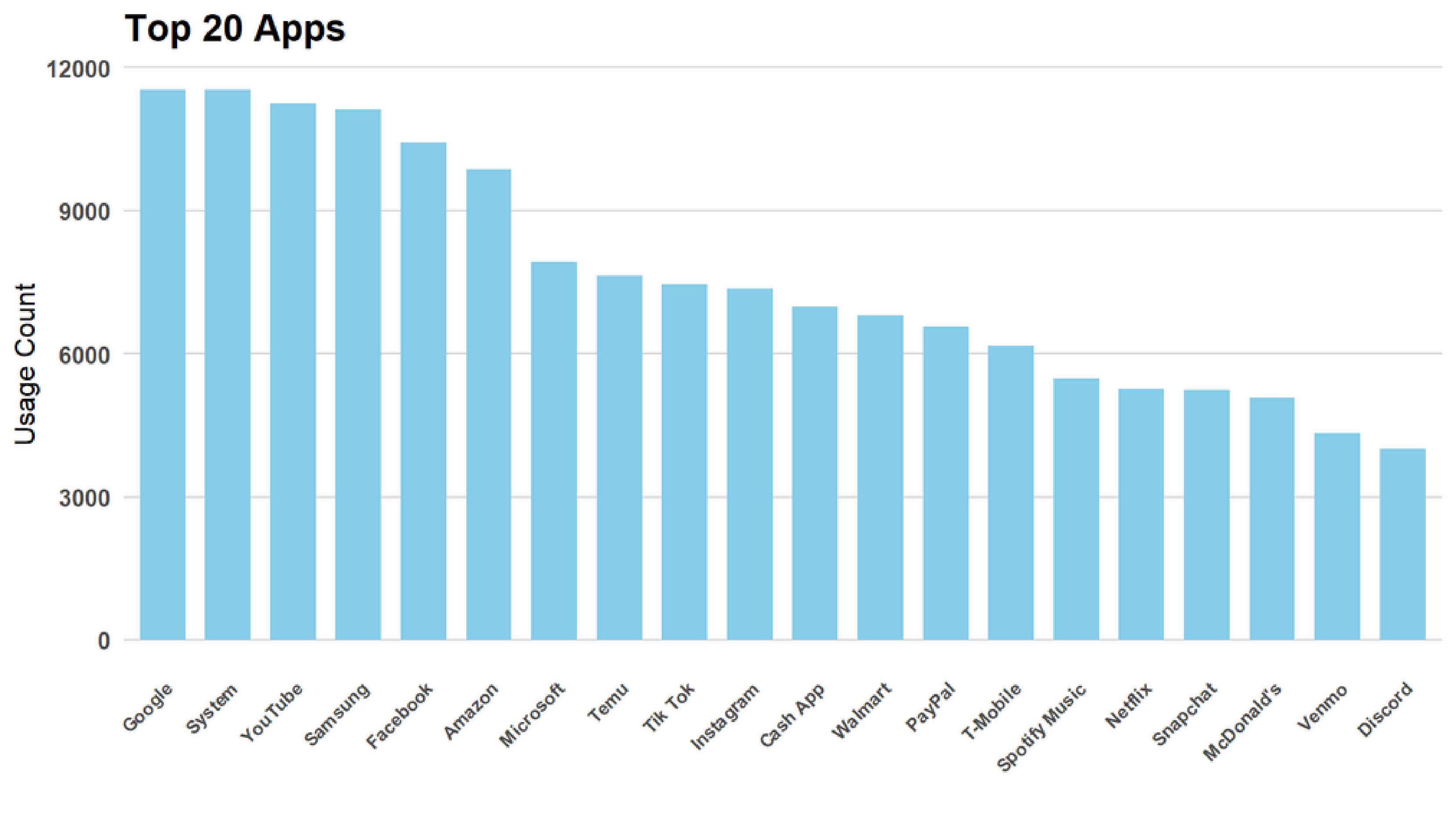
Which user segments does the use of generative AI applications affect through users' behavior and engagement across other mobile apps, and where does that shift open up business value?

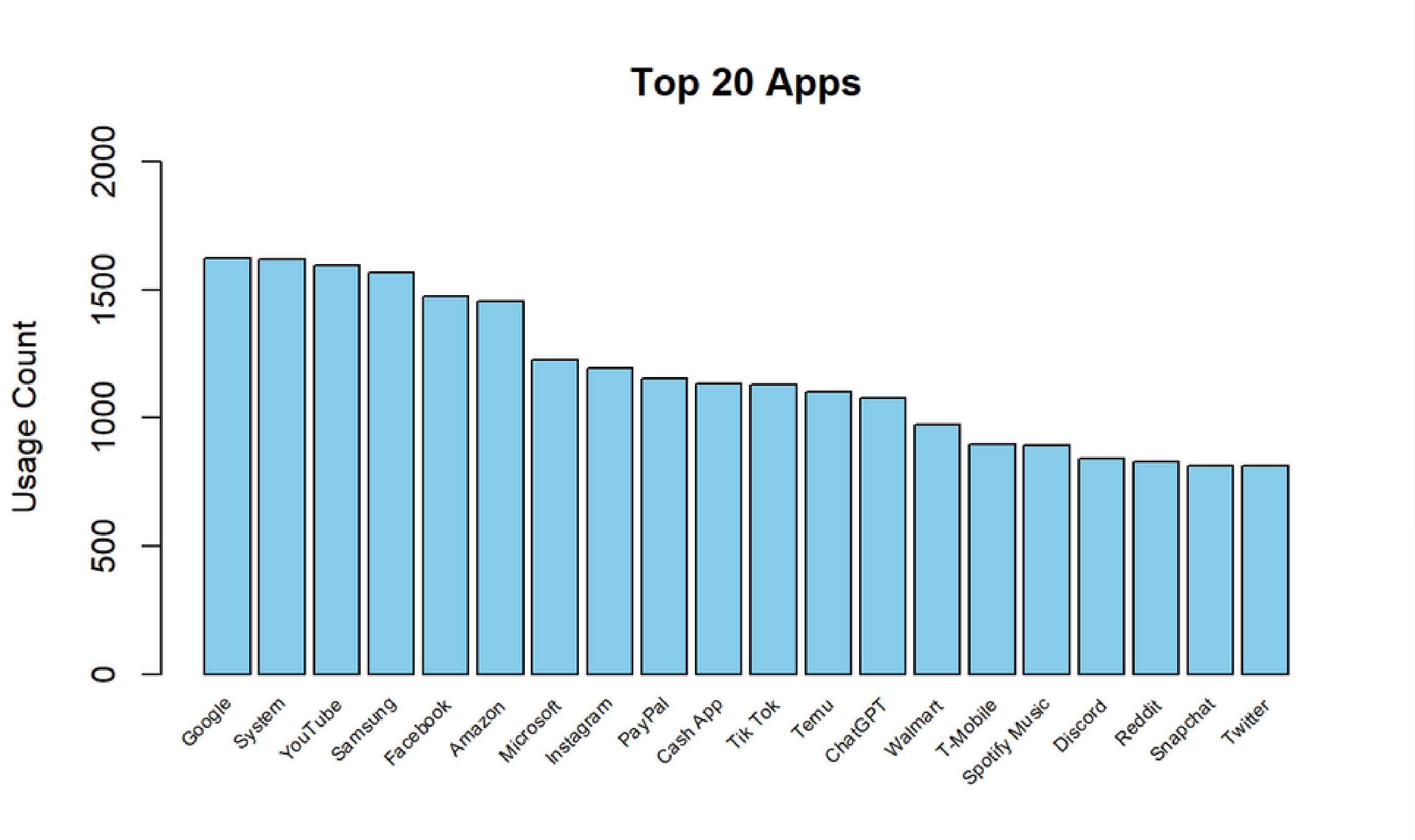
Initial Observations Before Association Rules: Insignificant Apps Dominate the Data Set and Mask Meaningful Patterns

System and large brand apps appear across almost every device and aren't tied to LLM behavior. This noise makes it difficult to intepret which apps actually correlate with LLM usage.

Raw App Frequencies

After Filtering





Association Rules: Isolating the Treatment Group and Setting Significant Parameters

The dataset's panel structure and scale require restructuring into transactions, where each transaction represents a panelist and each app functions as an item within that transaction, to enable Apriori rule mining.

Data Summary

The weekly usage dataset contains 20,647,355 observations, including a panelist ID, app description, and usage duration.

The data set follows a panel structure and includes 221,395 duplicate records and zero missing values.

In the data set, there are also inconsistent naming conventions. Ex: Coinbase and Coinbase.com.

Data Preparation

Our approach is as follows:

1. Convert usage duration into a binary variable
2. Encode app names for consistency
3. Collapse the panel structure to a single record per panelist, marking apps with a 1 if they were used
4. Group apps by panelist and filter those who've used a LLM
5. Create the transaction matrix using only the treated LLM users

AR Thresholds

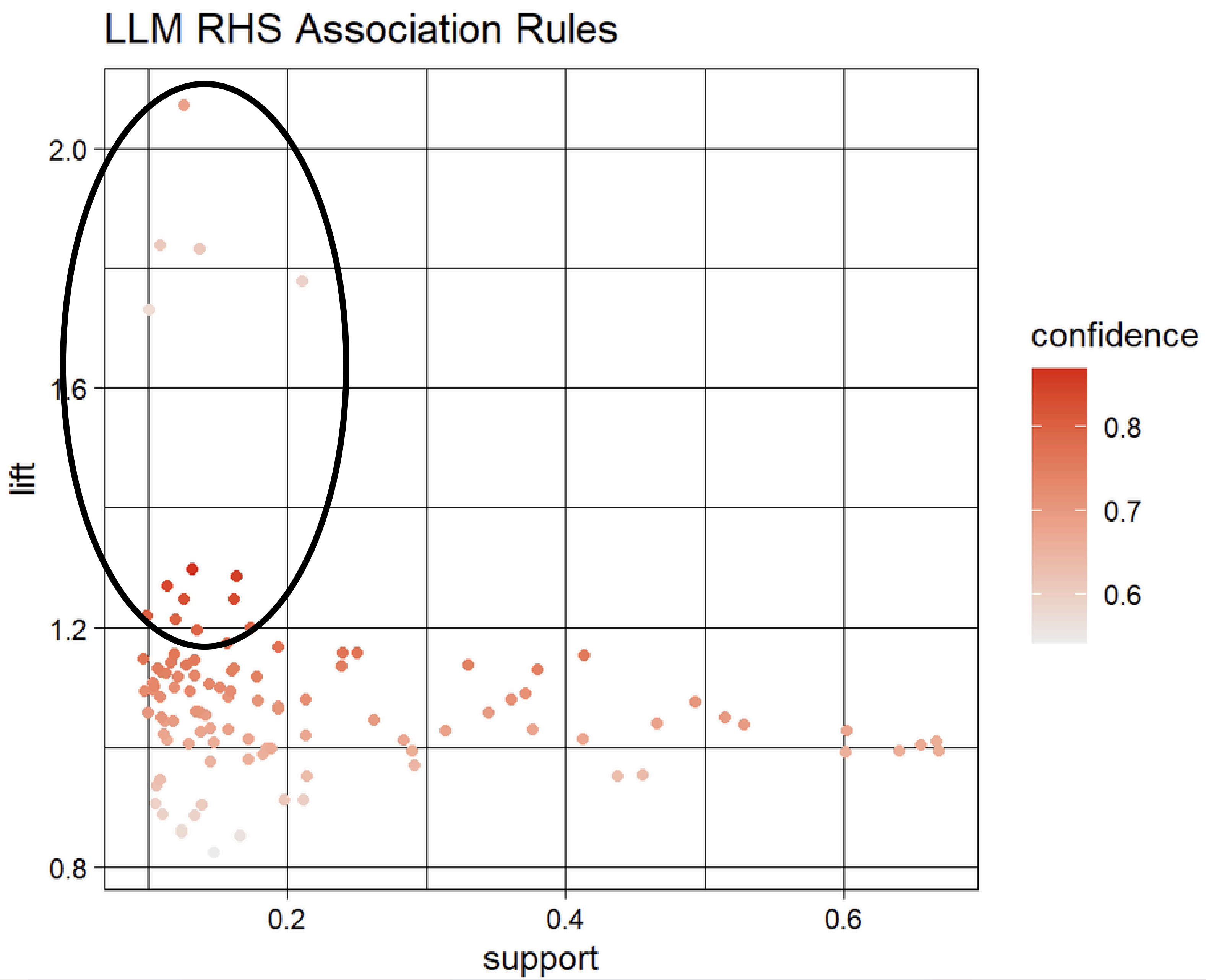
The following thresholds were selected for the Apriori algorithm:

- Support = 0.1
- Confidence = 0.5
- Min Length = 2
- Max Length = 2

After filtering for rules that included ChatGPT, Gemeni, or Perplexity, a total of 143 rules remained.

Association Rules: Targeting High-Lift LLM Adoption Patterns and Classifying User Segments

Applying a lift threshold greater than 1.2 to RHS rules highlights the strongest co-occurring app behaviors with LLMs, indicating user groups whose LLM usage appears at least 20% more frequently than those chosen at random.



1. Play Station → ChatGPT
2. Xbox → ChatGPT
3. Roblox → ChatGPT
4. Twitch → ChatGPT

Gamers

5. Etsy → ChatGPT
6. CapCut → ChatGPT
7. Threads → ChatGPT
8. Canva → ChatGPT

Digital Creators

9. Crypto.com → Gemini
10. Sofi → Gemini
11. Brave Browser → Gemini
12. Coinbase → Gemini

Investors

Association Rules: The Finance Co-Usage Pattern Stands Out as the Strongest Signal

Aggregating the outputs from association rules allows us to roughly quanitfy how strongly each user group's behavior exceeds baseline LLM adoption

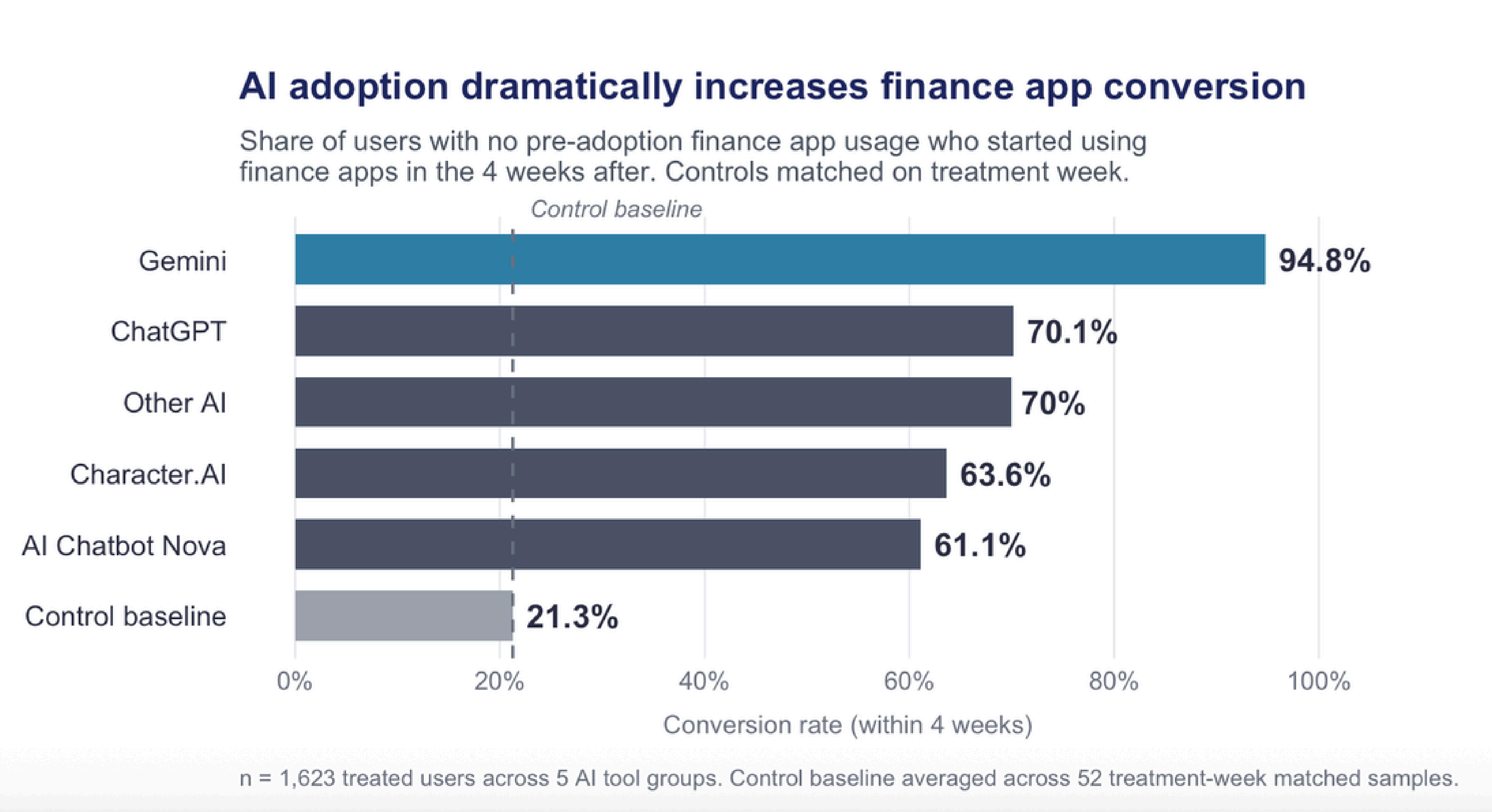
21% → **Gamers** represent 13.3 % of all LLM users and are 20.5% more likely to use ChatGPT than other groups selected at random.

30% → **Digital Creators** represent 14.5% of all LLM users and are 30% more likely to use ChatGPT than other groups selected at random

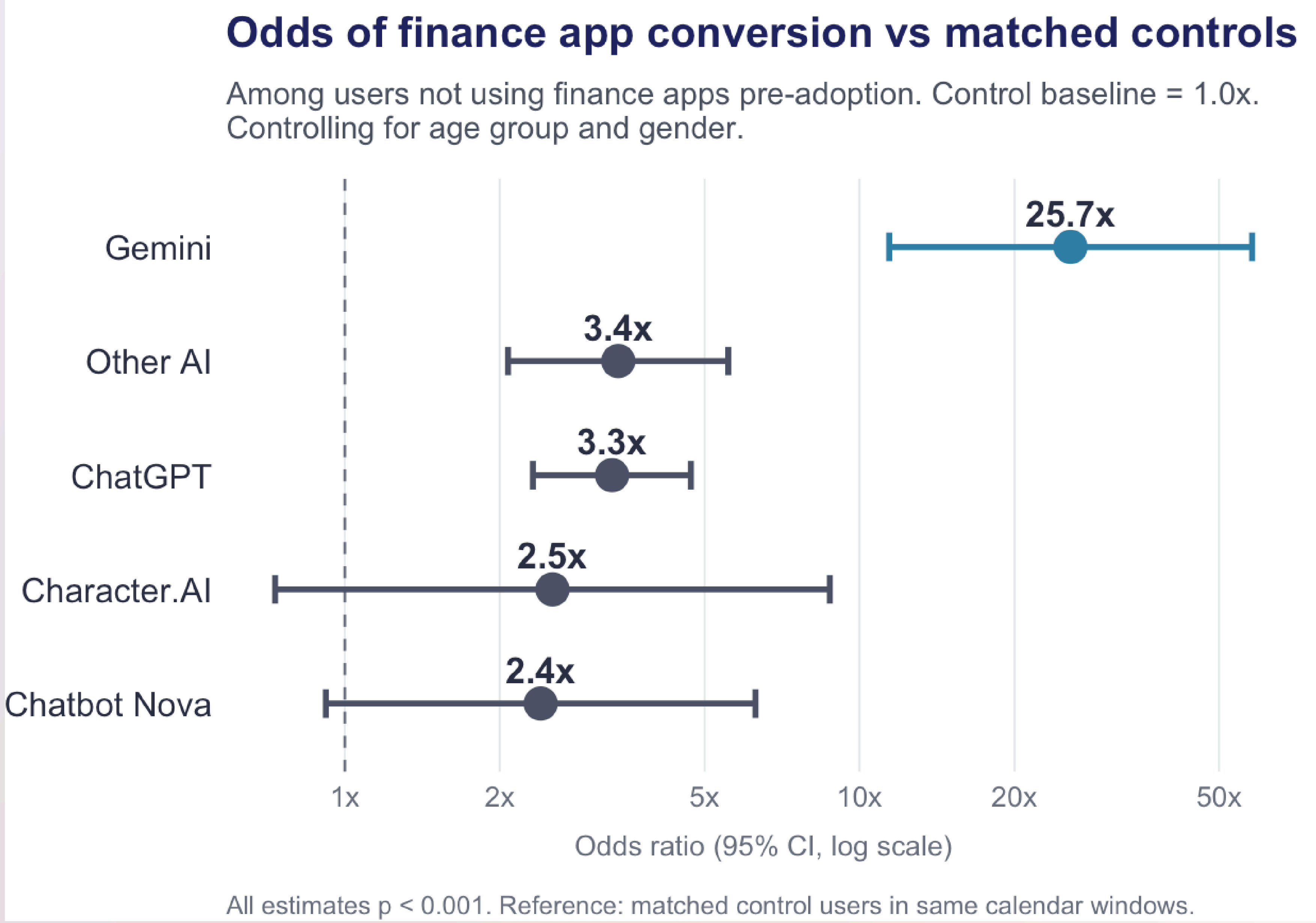
85% → **Investors** represent 13.7% of all LLM users and are 84.7% more likely to use Gemini than other groups selected at random

LLM Adoption Causally Drives Finance App Uptake

Among users with zero finance app activity in the four weeks before adopting AI, Gemini adopters converted at rates that matched control users did not reach.



Raw conversion rates — who became finance users after AI
Shifting from lift to conversion rates



Odds ratios after controlling for age and gender

95% of Gemini adopters who had never used finance apps before became finance users within 4 weeks of their first AI session. Matched controls in the same calendar weeks converted at just 21%.

METHODOLOGY

How We Arrived at the Causal Finding

Correlation was not enough. We built a four-step pipeline to test whether AI adoption actually changes what people do next on their phones.

01

Mark the treatment week

For each of 1,623 AI-adopting users, we found the earliest week any LLM app appeared in their usage. That became their personal treatment date.

02

Build pre and post windows

Looked at each user's finance app activity in the 4 weeks before their treatment week and the 4 weeks after. Equal windows on both sides.

03

Filter to true conversions

Kept only users with zero finance app activity in their pre-window. This strips out the selection story that AI users were already fintech people.

04

Match against controls

For the 52 unique treatment weeks, ran the same before-and-after test on non-AI users. Averaged to get the 21% baseline conversion rate.

WHY THIS DESIGN HOLDS UP

Matching on calendar weeks removes seasonality. Filtering on pre-window activity removes selection bias. Controlling for age and gender in the regression removes demographic confounding. What is left is the effect of AI adoption itself.

BUSINESS IMPACT

What a 25x Odds Ratio Actually Means for a Business

A behavioral signal this strong is worth money to anyone trying to reach high-intent financial customers. Here is where that value shows up.

TARGETING

~$700K

in potential efficiency savings per $1M quarterly campaign

Mobile finance ads run $15-$30 CPMs. Redirecting spend toward Gemini-adjacent users where conversion runs 4.5x baseline cuts effective acquisition cost proportionally.

PARTNERSHIPS

4+

crypto and neobroker platforms with strongest Gemini signal

Coinbase, SoFi, Robinhood, and Crypto.com could each build integrations or co-marketing around Gemini users. The behavioral overlap justifies platform-level deals.

PRODUCT

1

underpriced asset in Google's current monetization stack

Gemini itself is a natural surface for financial product discovery. Affiliate placements, native financial features, or direct broker integrations all become defensible.

The takeaway: Gemini adoption may be one of the strongest behavioral signals available in consumer mobile data for finding high-intent financial customers.

NEXT STEPS

Limitations & Future Work

What this analysis can and cannot tell you, and where we would take it next.

Limitations of This Study

- Single-year panel (2023 only), Android users
- Small treated subsamples for Character.AI (n=11) and Chatbot Nova (n=18)
- Gemini pre-installed on many Android devices, which may inflate adoption counts
- 4-week observation window may miss longer adoption cycles

What We Would Test Next

- Replicate on 2024 panel data to see if patterns hold
- Extend beyond finance to streaming, health, and productivity categories
- Add usage intensity (not just binary) as a covariate
- Test Claude, Copilot, and Perplexity as new AI tools mature

What a Practicioner Would Need

- Larger sample with iOS users for cross-platform validation
- Revenue or LTV data to attach dollar values to conversion signals
- A/B test on a real targeting campaign using Gemini-adjacent audiences
- Privacy-compliant data sourcing for production deployment

Recap and Thank You

What we built, what we found, and why it matters.

WHAT WE BUILT

A reproducible analytics pipeline

20.8 million mobile usage records, 442,790 duplicates cleaned, 14 app categories engineered, 5 AI canonical buckets. Association rules across 12 months paired with a matched pre/post regression.

WHAT WE FOUND

A causal signal, not just a correlation

Gemini users with no prior finance app activity became finance users at 25.7 times the rate of matched controls. The strongest single behavioral signal in the dataset, and one of the strongest in consumer mobile data overall.

WHY IT MATTERS

AI adoption is a targeting asset

Financial services companies can cut acquisition costs dramatically by using AI tool adoption data in their customer models. Google can monetize Gemini's audience more directly. The methodology generalizes as new AI tools emerge.

"AI adoption is a behavioral signal. The question is who acts on it first."

Team 7 | Cheeseng Her | Thomas Kim | Siyu Liang | Shreya Muthukumar

Questions welcome.